Convergence Claims for the CCM Construction: An Empirical Reassessment

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Version 2.1

Abstract

I revisit the three experiments in my earlier paper on the Connes–Consani–Moscovici (CCM) construction: a Weil–prolate comparison, a test motivated by Śliwiński's spectral-error conjecture, and auxiliary Mellin-transform comparisons. Corrected calculations and a review of the quantities being compared change the conclusions. The Weil–prolate experiment shows close finite sampled agreement and does not test CCM Lemma 7.2. The spectral experiment now examines maximum error along the prescribed scaling $\lambda = N$. Independent interval checks account for all 1700 positive roots in seven saved finite spectra. Across a tenfold increase in the parameter, maximum error grows about 49-fold; growth persists after excluding the highest 20% of roots. A coefficient-based trace bound independently excludes small maximum errors in these saved spectra. The same calculations give increasingly accurate initial roots. This separation between initial-root accuracy and expanding-window error is evidence against the proposed uniform inverse-logarithmic decay in the tested range. Analytic source-error control and an asymptotic disproof remain open. The Mellin scans located real-part crossings that fail the complex-zero check. A separate boundary argument proves that the tested fixed-cutoff surrogates have only finitely many critical-line zeros, although the untruncated weight represents zeta exactly. This identifies a truncation obstruction without challenging CCM's own transform. I withdraw the earlier falsification claims and retain the corrected measurements, their limitations, and the reproduction scripts.

1 Purpose of this correction

The earlier version of this paper interpreted three numerical tests as evidence against proposed explanations of the accuracy of the CCM construction [1]. This revision addresses those same tests. It corrects the attribution, numerical results, and conclusions. Its stance on the spectral-error conjecture is empirical: the seven finite comparisons give an adverse trend in the specified maximum-error observable. This is evidence against the proposed decay in the tested range, not a proof about the limit.

I apologize to the CCM authors for mislabeling the Weil–prolate comparison as a test of their Lemma 7.2. That lemma compares suitably normalized prolate and Hermite functions. The approximation of a Weil state by a prolate trial is a different question. I withdraw the claim that my experiment falsifies Lemma 7.2.

Two further corrections are necessary. Śliwiński's Conjecture 4.1 concerns maximum error, whereas the earlier rate argument used mean error [2]. The auxiliary Mellin experiments located zeros of a real part, not validated complex zeros. Neither comparison supports the earlier conclusions in the form stated. The following sections report what the retained evidence does support.

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2 Numerical conventions

Write $C = \lambda^2$ and let N be the positive Fourier cutoff. The finite Weil matrix has dimension $2N + 1$; its even sector has dimension $N + 1$. The experiments use a retained even coefficient vector, the associated finite Weil function, and, where requested, the positive movable roots of its secular expression. These finite objects are not identified here with the limiting CCM state. Below I identify the complete finite root list with the saved finite rank-one spectrum; the full infinite spectrum is a different object.

The revised runs use arbitrary-precision arithmetic through the Xcelerator Toolkit, with source identities, numerical settings, and software revisions recorded in each journal. The letter P denotes requested decimal working precision, not certified output digits. The $P = 1000$ and $P = 2000$ sources below use 3386 and 6708 bits, respectively; numerical kernels may increase precision according to their error checks. Historical outputs retain their original settings.

The current root-acquisition route is independent of the reference zeta ordinates; the subsequent comparisons use a retained reference dataset. Successful execution, finite hypothesis outcomes, and supplemental diagnostic coverage are recorded separately. A completed experiment can have a negative result or leave an asymptotic question unanswered.

3 Test 1: the Weil–prolate comparison

The trial is constructed from a zero-integral combination of the even prolate modes with zero and four nodes, followed by the finite sum map

$$h_\lambda = h_{4,\lambda} - r_\lambda h_{0,\lambda}, \qquad r_\lambda = \frac{\int h_{4,\lambda}(x)\,dx}{\int h_{0,\lambda}(x)\,dx}, \qquad k_\lambda(u) = \sqrt{u} \sum_{1 \le n \le \lambda/u} h_\lambda(nu).$$

The numerical implementation uses a three-point finite-difference prolate model with zero endpoint values and piecewise-linear interpolation. It is a finite-Dirichlet approximation, not a certified evaluation of the regular continuum prolate functions.

For the 1024 logarithmically spaced retained samples, let x_i denote the finite Weil function and g_i the trial. The reported observable is the sampled maximum residual after least-squares scaling:

$$a_{\mathrm{LS}} = \frac{\sum_i x_i g_i}{\sum_i g_i^2}, \qquad E_{\mathrm{LS}} = \frac{\max_i |x_i - a_{\mathrm{LS}} g_i|}{\max_i |x_i|}.$$

It is neither a continuous supremum nor the minimax fit over scalars.

Table 1: Earlier reported residuals and current sampled residuals. The earlier rows used $P = 1000$; P below is the current setting. Source policy and implementation also changed.

Claim	C	N	Grid	P	Earlier	Current E_{LS}
1a	13	120	4001	1000	6.887×10^{-5}	$6.88681023 \times 10^{-5}$
1b	100	500	4001	1000	6.697×10^{-4}	$6.71252770 \times 10^{-4}$
1c	100	500	8001	1000	6.470×10^{-4}	$6.46956953 \times 10^{-4}$
1d	1000	800	8001	2000	9.988×10^{-1}	$7.08120048 \times 10^{-3}$

The current fits have been independently replayed on the serialized sample values. At $C = 1000$, the residual is approximately 0.7081%, replacing the earlier 99.88%. The description of the profiles as essentially uncorrelated is therefore withdrawn. The change cannot be attributed to precision alone. At $C = 100$, the two grid sizes share the same Weil-source identities; grid refinement lowers the residual by about 3.62%. Two grids do not establish the continuum error or remove the need to control the finite-difference model.

CCM Lemma 7.2 gives an absolute uniform prolate–Hermite estimate, after normalization. It is not a relative Weil–prolate error bound. Moreover, finitely many values of CE_{LS}, with changing Fourier cutoff and numerical resolution, cannot disprove an asymptotic bound with unspecified constant and onset. The defensible result is close sampled agreement for these four finite configurations; the asserted falsification does not survive.

The missing transfer estimate can be made explicit. Denote the preceding sum map by T_λ. On $\lambda^{-1} \le u \le \lambda$, the triangle inequality gives

$$|T_\lambda h(u)| \le \frac{\lambda}{\sqrt{u}} \|h\|_\infty, \qquad \|T_\lambda h\|_\infty \le \lambda^{3/2} \|h\|_\infty. \tag{1}$$

Even an absolute input error $O(\lambda^{-2})$ therefore yields only the crude transferred bound $O(\lambda^{-1/2})$ by this argument. This is an upper bound, not a prediction of the actual decay; cancellation could improve it. A relative Weil–prolate estimate additionally needs control of the Weil approximation and its normalization. Dunster's uniform prolate asymptotics provide error bounds in specified parameter regimes [3]; applying such results here would still require matching those regimes and controlling the finite-Dirichlet discretization. This identifies what a genuine convergence test would have to measure.

4 Test 2: the spectral-error comparison

Let $\mu_1 < \cdots < \mu_N$ be the complete positive movable-root list of a retained finite source and let γ_k be the kth positive reference zeta ordinate. The experiment records both

$$A_N = \frac{1}{N} \sum_{k=1}^{N} |\mu_k - \gamma_k|, \qquad B_N = \max_{1 \le k \le N} |\mu_k - \gamma_k|.$$

It also retains a separately labeled trimmed mean. These are different observables and are not interchanged here.

Dominik Śliwiński clarified in correspondence that the intended observable is maximum (L^∞) error, rather than mean error. His clarification prompted me to re-evaluate this test using maximum error.

His published Definitions 2.5–2.6 distinguish mean error ϵ from maximum error $\mathcal{E}$. Conjecture 4.1 asserts the existence of $\lim_{\kappa\to\infty} \mathcal{E}(\kappa) \log \kappa$ along $\lambda = N = \kappa$. A nonzero limit is suggested separately after the displayed conjecture. The earlier paper substituted the mean and treated the nonzero value as part of the conjecture. Those statements are corrected here.

Crucially, Definition 2.6 maximizes over the N positive eigenvalues of the finite $(2N+1)$-dimensional operator in Definition 2.4. For a saved even vector w, put $S = \sum_j w_j \neq 0$, let $D = \mathrm{diag}(d_j)$ be the Fourier diagonal, and set $A = D - (Dw)\mathbf{1}^T/S$. The rank-one determinant identity gives

$$\det(zI - A) = \frac{z}{S} Q(z), \qquad Q(z) = \prod_{j=-N}^{N} (z - d_j) \sum_{j=-N}^{N} \frac{w_j}{z - d_j}.$$

This is the finite construction underlying CCM Lemma 5.4. For these saved sources all weights are nonzero. Independent directed interval checks bracket N distinct positive roots away from the poles; symmetry supplies N negative partners, exhausting the degree $2N$ of Q. Thus the checked roots are the complete positive spectrum of the saved finite matrix A. The unchanged lattice modes of the infinite CCM operator do not belong in this finite comparison. A full-infinite-spectrum indexing caveat would address a different observable.

The seven scripts 2a–2g have been run at $\kappa = 50, 100, 150, 200, 300, 400, 500$, with $C = \kappa^2$. The completed runs use $P = 1000$ through $\kappa = 300$, $P = 2000$ at $\kappa = 400$, and $P = 1500$

at $\kappa = 500$. Required measurements and applicable supplemental acquisitions are accounted for through original runs and retained-source recoveries. Table 2 reports all seven current maximum-error rows. The independent Rust verifier uses directed MPFR intervals and no Toolkit numerical code. It checks all 1700 roots and computes errors against the pinned reference dataset. Original journals support the first five rows; authenticated published source/root artifacts support the last two. The brackets certify the exact saved sources, not their distance from the analytically defined Weil ground states.

Table 2: Complete saved finite spectra on $\lambda = N = \kappa$, $C = \kappa^2$. The full maximum B_N occurs at $k = N$ in every row. $B_{0.8N}$ is the maximum over $1 \le k \le \lfloor 0.8N \rfloor$, excluding the highest 20% of roots.

$\kappa = N$	P	B_N	$B_N \log \kappa$	$B_{0.8N}$
50	1000	3.85480×10^3	1.50800×10^4	54.4129
100	1000	1.20360×10^4	5.54278×10^4	87.9902
150	1000	2.36891×10^4	1.18697×10^5	112.208
200	1000	3.87289×10^4	2.05198×10^5	134.727
300	1000	7.77216×10^4	4.43307×10^5	177.449
400	2000	1.28071×10^5	7.67333×10^5	216.408
500	1500	1.89365×10^5	1.17683×10^6	254.015

Every successive row has a larger full maximum and a larger logarithmically scaled maximum. Across the tenfold parameter increase, these quantities grow about 49-fold and 78-fold, respectively. The effect is not confined to the final root: excluding only that root leaves maxima increasing from 1106.21 to 60077.4; excluding the highest 20% still leaves a 4.67-fold increase. Conversely, the maximum over the first half of each list falls from 8.43×10^{-6} to 2.48×10^{-69}. These computations separate improving low-lying accuracy from worsening error over an expanding window.

A check independent of root refinement

The size of the full-window discrepancy can also be bounded from the coefficients alone. Expanding the square of the finite matrix A gives

$$\operatorname{tr}(A^2) = \sum_j d_j^2 - \frac{2\sum_j w_j d_j^2}{S} + \left(\frac{\sum_j w_j d_j}{S}\right)^2 .$$

Even weights and antisymmetric poles make the last term zero. The complete spectrum $0, \pm\mu_1, \ldots, \pm\mu_N$ therefore satisfies

$$M_2 := \sum_{k=1}^{N} \mu_k^2 = \frac{1}{2}\operatorname{tr}(A^2) = \sum_{j=1}^{N} d_j^2 - \frac{\sum_{j=-N}^{N} w_j d_j^2}{S}. \tag{2}$$

This trace identity does not require the displayed coordinate matrix to be normal. The reverse triangle inequality for the Euclidean norm, together with $\|\mu - \gamma\|_2 \le \sqrt{N} B_N$, yields

$$B_N \ge \max\left\{0, \frac{\sqrt{M_2} - \sqrt{\sum_{k=1}^{N} \gamma_k^2}}{\sqrt{N}}\right\}. \tag{3}$$

I evaluated these bounds locally with independent directed MPFR intervals, using the exact saved coefficients and the pinned reference ordinates rounded to each source precision. No root values enter (3). The separate sum of squared verified root intervals overlaps the coefficient enclosure in all seven rows. Thus the large discrepancies are not explained by root refinement

Table 3: Conservative lower bounds from coefficients alone, using (3). Integers are rounded downward. The separate root-bracket and coefficient enclosures for M_2 overlap in every row.

N	50	100	150	200	300	400	500
Lower bound on B_N	525	1196	1952	2792	4633	6663	8860

alone. This check still depends on the saved ground-state coefficients: it does not certify their error relative to the analytic construction. In particular, the weights are signed, so the ratio in (2) is not a positive weighted average of d_j^2. Large outlying roots cannot be rejected by an interlacing argument that assumes positive weights.

Interpretation and remaining scope

This is substantive evidence against inverse-logarithmic maximum-error decay in the tested range, including the suggested faster powers of the logarithm: the observed maximum increases rather than decreases. It does not establish the eventual behavior. The remaining numerical obligation is to propagate matrix-assembly and ground-state uncertainty to the spectral error, supported by independent precision refinements. An asymptotic disproof additionally needs an eventual lower bound or another analytic contradiction. The new maximum-error table replaces the earlier mean-based rate argument; it does not restore that argument.

There is also a window issue in the published supporting evidence: Section 4 of [2] reports comparisons with the first 1000 zeta zeros while extending $N = \kappa$ to 7500. For $N > 1000$, that reported reference window alone cannot establish the maximum over all N eigenvalues in Definition 2.6. A fixed window and an expanding all-N window require separate evidence.

Recent work also illustrates the strength of the initial-root phenomenon. Groskin reports 307–329 matching digits for the first ten zeros in a truncated-Weil calculation at his cutoff $c = 100$, $N = 250$ [4]. That is a different finite configuration, and I do not independently validate those numbers here. Such initial-window accuracy is compatible with the separation observed above; it does not measure the all-N maximum along $\lambda = N$.

Two additional standard configurations, which are not on $\lambda = N$, now have independently checked complete movable-root lists and error arithmetic:

Table 4: Correction of two historical standard-row means. Current rows use $P = 1000$. These are full movable-window statistics, not a matched test of Śliwiński's asymptotic conjecture.

C	N	Earlier mean	Current A_N	Current B_N
13	120	1.063	26.61079659	1862.704134
100	500	1.595	427.60878966	86620.999761

Independent finite matrix/vector checks and complete stored-source root checks support the current rows. The historical route used reference-seeded refinement, whereas the present route discovers the complete positive movable-root list. The original per-root outputs needed to establish the precise cause of the discrepancies were not recovered. The earlier values must not be presented as reproduced current full-window means.

The distinction between fixed-index convergence and a uniform all-window rate is central to the adverse result above. The earlier claim of super-exponential convergence is also withdrawn: linear growth of matching digits alone would indicate exponential decay in that parameter, not super-exponential decay.

5 Test 3: the auxiliary Mellin transforms

The two auxiliary transforms tested were

$$\omega(u) = \frac{ue^u}{(1+e^u)^2}, \qquad F_\lambda(s) = \int_{\lambda^{-1}}^{\lambda} \omega(u) u^{s-1}\, du, \qquad G_{C,N}(s) = \int_{\lambda^{-1}}^{\lambda} f_{C,N}(u)\omega(u) u^{s-1}\, du,$$

where $f_{C,N}$ is the finite Weil function. The scripts call these the naive and weighted transforms. They scan the real part on $s = 1/2 + it$, for $5 \le t \le 55$, using 5000 intervals and 30 requested refinement digits. The retained evaluations use $P = 1000$.

A real-part crossing is not sufficient for a complex zero. The revised checks evaluate both components at quadrature orders 500 and 1000, normalized by the corresponding absolute-weight integral, with residual threshold 10^{-20}.

Table 5: Observed real-part crossings and failed complex-zero checks. The scans do not certify completeness of the crossing or zero sets.

Claim	C	Transform	Crossings	Failed checks
3a	13	Naive	21	21
3a	13	Weighted ($N = 120$)	11	11
3b	100	Naive	37	37

The quadrature-refinement checks passed, but every listed crossing failed the complex residual criterion. Thus the earlier accuracy ratio between these crossings and CCM roots compared different mathematical objects and is withdrawn. The negative tests concern these observed crossings; they do not prove that the transforms have no other complex zeros.

A fixed-cutoff obstruction

The untruncated weight is a classical zeta representation [5]:

$$F_\infty(s) = \int_0^\infty u^s \frac{e^u}{(1+e^u)^2}\, du = \Gamma(s+1)(1-2^{1-s})\zeta(s), \qquad \Re s > 0. \tag{4}$$

The apparent singularity at $s = 1$ is removable. On $\Re s = 1/2$ both prefactors are nonzero, so this integral has exactly the zeta zeros on that line. Hard truncation changes that conclusion.

Proposition (fixed-cutoff obstruction). At $C = 13$ and $C = 100$, F_λ has no critical-line zeros for $|t| > 100$ and $|t| > 2000$, respectively. For either fixed cutoff, the naive and nonzero finite cosine-weighted transforms have only finitely many critical-line zeros.

Proof. Set $L = \log\lambda$, $h(u) = e^u/(1+e^u)^2$, and $q(x) = e^{3x/2}h(e^x)$. The naive transform is $\int_{-L}^{L} q(x)e^{itx}\, dx$. Two integrations by parts give

$$|F_\lambda(1/2+it)| \ge \frac{\delta}{|t|} - \frac{M}{t^2}, \quad \delta = |q(L) - q(-L)|, \quad M = |q'(L)| + |q'(-L)| + \int_{-L}^{L} |q''(x)|\, dx. \tag{5}$$

Writing $u = e^x$ and $r = \tanh(u/2)$ gives $q' = q(3/2 - ur)$ and $q'' = q((3/2-ur)^2 - ur - u^2(1-r^2)/2)$. Since $0 < r < 1$ and $h(u) \le e^{-u}$,

$$M \le B_\lambda := q(-L)(3/2 + \lambda^{-1}) + q(L)(3/2+\lambda) + \frac{111\sqrt{\pi}}{16}.$$

The last term integrates the upper bound $u^{1/2}e^{-u}(9/4 + 4u + 3u^2/2)$ over $(0,\infty)$. Outward rounding at both 256 and 512 bits gives

C	enclosure for δ	B_λ/δ upper bound
13	[0.14049, 0.14050]	95
100	[0.006450, 0.006451]	1911

so (5) proves the stated conservative thresholds.

For $p(x) = f_{C,N}(e^x)$, a nonzero finite cosine polynomial in $j\pi x/L$, endpoint derivatives agree and odd ones vanish. Analyticity gives a first nonzero endpoint derivative of finite even order k. The lower endpoint derivatives of pq vanish. Integrating $k+2$ times therefore gives a leading modulus at least $|p^{(k)}(L)|\delta/|t|^{k+1}$ and remainder $O(|t|^{-k-2})$. Both transforms are eventually nonzero, entire, and not identically zero; only finitely many critical-line zeros remain. □

These fixed surrogates cannot reproduce the unbounded sequence of zeta critical-line zeros. This does not exclude off-line zeros, certify the entire scanned interval, quantify a weighted threshold, or address λ or N growing with height.

What increasing the cutoff does recover

The obstruction is compatible with convergence of the transform values. For every $\lambda > 1$ and real t, the omitted tails obey

$$|F_\infty(1/2+it) - F_\lambda(1/2+it)| \le \frac{1}{6\lambda^{3/2}} + \sqrt{\lambda}\, e^{-\lambda}\left(1 + \frac{1}{2\lambda}\right). \tag{6}$$

Indeed, $h(u) \le 1/4$ bounds the lower tail by $\int_0^{1/\lambda} u^{1/2}/4\, du = 1/(6\lambda^{3/2})$. For the upper tail, use $h(u) \le e^{-u}$ and the concavity bound $\sqrt{\lambda+v} \le \sqrt{\lambda} + v/(2\sqrt{\lambda})$, then integrate against e^{-v} for $v \ge 0$. Neither estimate depends on t. For example, the right-hand side is less than 0.005422 at $\lambda = 10$, and less than 0.000167 at $\lambda = 100$.

At any critical-line zeta zero, (4) makes the full integral zero, so (6) bounds the truncated integral's residual there. A small residual does not force an exact zero on the same line. Hence two facts coexist: increasing λ recovers the full transform uniformly in absolute value on that line, while fixing either tested cutoff eventually excludes critical-line zeros. The bound gives a sufficient accuracy budget for transform values; it does not control zero locations. Such control would require, for example, a contour estimate comparing the tail with the full transform away from its zeros.

CCM Proposition 5.9 and Theorem 5.10 use a different multiplicative Fourier transform representing the secular spectral information under their hypotheses. Claim 3a's same-source Fourier/secular control passed. Neither this obstruction nor the crossing tests support the earlier claim that CCM accuracy cannot be explained by an integral transform.

6 Conclusion

The revised results show close finite Weil–prolate agreement and a definite obstruction for the tested fixed Mellin cutoffs, whose untruncated counterpart represents zeta exactly. All seven checked finite spectra give an adverse trend in the specified expanding-window maximum, even after excluding the highest 20% of roots. The coefficient-based moment bounds independently rule out small full-window errors for those same saved sources. The proposed uniform decay is empirically unsupported in this range. Independent analytic source control and an eventual lower bound are the next steps toward a stronger conclusion.

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Statement on the Use of AI Tools

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Data and code availability

The source repository provides the scripts and `docs/manuscript-results.json` binding these tables to retained evidence. Full audit payloads are maintained separately. This version revises the earlier published paper.

References

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